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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/17/2006 |
| **PI Name:** | Banish, R. M. Ph.D. | | |
| **Project Title:** | Self-Diffusion in Liquid Elements | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Basics of solidification | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 35899 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | TRANSITIONAL (2004-2007) | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 09/01/1996 | **End Date:** | 09/01/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Alexander, John I.D. ( Case Western Reserve University ) | | |
| **Grant/Contract No.:** | NCC8-99 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

Toward the task objective we pursued the following scientific tasks:
- Measurements of diffusivities of selected materials that will be chosen according to class-like molecular interaction behavior in the liquid.

- Investigation of the significance of the "wall effect" in diffusion capillaries.

- Exploration of convective contamination of diffusivity measurements on Earth through numerical modeling; combined with our low-gravity determinations.

Toward completing the above scientific tasks we are pursuing the following hardware development goals:

- Development of an efficient technique for dynamic in-situ measurements of diffusivities in melts as a function of

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| **Task Description:** | temperature.<br><br>- Development of a flight-certified hardware package to automatically perform such diffusivity measurements under reduced gravity conditions and on Earth.<br><br>Diffusivities will be determined from temporal records of evolving concentration profiles through measurement of radioactive tracer emission. An initially solid, cylindrical sample contains a radioactive isotope at one end. After melting, radiation escaping through small bores in an isothermal liner/radiation shield is monitored via two advantageously placed detectors. Data evaluation is facilitated by an algorithm which is not limited to the simple initial conditions traditionally used in diffusivity studies. The algorithm permits data deduction from any sequence of concentration distributions. Hence, diffusivity data can be gathered over a range of temperatures in a single experiment. Utilizing the different radiation absorption behavior of the different photon energies, we investigated the significance of the "wall effect" in indium metal. The effect is currently believed to contaminate diffusion studies in narrow capillaries used to suppress convection at normal gravity. Results from our low-gravity flight experiment during NASA/MIR (as a risk mitigation experiment), see Liquid Metal Diffusion project, indicated that there is no "wall effect" in indium (using a nonwetting container).<br><br>This research aims at identifying the temperature dependence(ies) and diffusion mechanisms of self-diffusion for liquid metals with different liquid structures. These dependencies will be determined via real-time, in-situ, self-diffusion measurements over a wide range of temperatures. Both benchmark data sets for industrial/scientific uses close to the melting point, i.e., during solidification/casting, and, wide-temperature range data for liquid theory elucidation will be acquired. At the present time there are eight different temperature-dependence predictions for self-diffusivion. Typical experimental diffusivity values vary by tens to hundreds of percent. Any extension to complex alloy systems for fundamental understanding and industrial use is strictly based on "best-guesses."<br><br>Understanding the liquid state structure and its influence on materials processing is important both for insight into the fundamental physics of liquids as well as for high fidelity metallurgical process control algorithms and prediction. The liquids structure and its change with temperature effects the transport of mass and heat through the system. Presently there are eight different predictions of the temperature dependence for self-diffusivity. These predictions are based on different assumptions of the liquids structure and the interchange of particles. Presently available self-diffusivity data sets vary between 10’s and 100’s of percent. Thus, any extrapolation of available terrestrial data to a unique temperature dependence is a guess. In addition to a better fundamental understanding of liquids the diffusion of species in melts and its temperature dependence is important for product quality in numerous metallurgical and semiconductor manufacturing processes. Hence, a detailed understanding of diffusion in liquid metals and alloys is essential for an efficient measurement of numerous technological processes. The uncertainties in the experimental values are due to difficult-to-control transport contributions from convection. These convective contaminations present in essentially all ground based measurements are due to the unavoidable (small) temperature nonuniformities that are present in molten systems. These small temperature nonuniformities lead to buoyancy driven convection which then contaminates the diffusivity determination by possibly 100’s of percent. This investigation will provide accurate data for selected liquids (molten elements) over wide temperature ranges that can be used for both process development and verification of theoretical models. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | No progress provided this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |